

5. 線形代数の基礎

$$(1) \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 + 8 = \underline{11}_{\#}$$

$$(2) \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 + 12 \\ 15 + 24 \end{bmatrix}$$

$$= \underline{\begin{bmatrix} 17 \\ 39 \end{bmatrix}}_{\#}$$

$$(3) \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 9 + 4 \\ 15 + 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 13 \\ 21 \end{bmatrix}$$

$$= 13 + 42$$

$$= \underline{55}_{\#}$$

演習問題は
(問4-1)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad f_1'(x) \text{ と } f_2'(x) \text{ を求めよ。}$$

$$f_1(x) = x^3$$

$$f_1'(x) = \lim_{h \rightarrow 0} \frac{f_1(x+h) - f_1(x)}{h} \quad \text{①}$$

$$\begin{aligned} f_1(x+h) &= (x+h)^3 \\ &= x^3 + h^3 + 3xh(x+h) \\ &= x^3 + h^3 + 3x^2h + 3xh^2 \end{aligned}$$

$$\text{①} = \lim_{h \rightarrow 0} \frac{x^3 + h^3 + 3x^2h + 3xh^2 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(h^2 + 3xh + 3x^2)}{h}$$

$$= \lim_{h \rightarrow 0} (h^2 + 3xh + 3x^2) = 3x^2$$

$$\underline{f_1'(x) = 3x^2}$$

$$f_2(x) = \sqrt{x}$$

$$f_2'(x) = \lim_{h \rightarrow 0} \frac{f_2(x+h) - f_2(x)}{h} \quad \text{②}$$

$$f_2(x+h) = \sqrt{x+h}$$

$$\begin{aligned}
 ② &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{(\sqrt{x+h} + \sqrt{x})}{(\sqrt{x+h} + \sqrt{x})} \\
 &= \lim_{h \rightarrow 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \\
 &\quad f'_2(x) = \frac{1}{2\sqrt{x}} \quad \#
 \end{aligned}
 \quad \left[\begin{array}{l} \text{確め算} \\ \sqrt{x} = x^{\frac{1}{2}} \\ = \frac{1}{2} (x)^{-1} \\ = \frac{1}{2} x^{-\frac{1}{2}} \end{array} \right]$$

(P2) 4.2)

$$(cf(x))' = cf'(x), (f(x) + g(x))' = f'(x) + g'(x)$$

$$f_1(x) = 2x^2 - 5x + 1$$

$$\begin{aligned}
 f'_1(x) &= (2x^2 - 5x + 1)' \\
 &= (2x^2)' + (-5x)' + (1)' \\
 &= 2(x^2)' - 5(x)' + 0 \\
 &= 2 \cdot 2x - 5 \cdot 1 \\
 &= 4x - 5
 \end{aligned}$$

$$f'_1(x) = 4x - 5 \quad \#$$

$$f_2(x) = \sum_{k=1}^{123} (k^2 + 1)x^k$$

$$\begin{aligned}
 f'_2(x) &= \left(\sum_{k=1}^{123} (k^2 + 1)x^k \right)' \\
 &= \sum_{k=1}^{123} ((k^2 + 1)x^k)' \\
 &= \sum_{k=1}^{123} (k^2 + 1)(x^k)'
 \end{aligned}$$

$$= \sum_{k=1}^{123} (k^2 + 1)k x^{k-1}$$

$$f'_2(x) = \sum_{k=1}^{123} (k^2 + 1)k x^{k-1} \quad \#$$

(PA 4.3)

$$(f(x) \cdot g(x))' = f'(x)g(x) + f(x) \cdot g'(x)$$

$$f_1(x) = x \cdot x$$

$$\begin{aligned} f_1'(x) &= (x)' \cdot x + x \cdot (x)' \\ &= x + x \\ &= 2x \end{aligned}$$

$$\underline{f_1'(x) = 2x_{\#}}$$

$$f_2(x) = x \cdot x^2$$

$$\begin{aligned} f_2'(x) &= (x)' \cdot x^2 + x \cdot (x^2)' \\ &= x^2 + 2x^2 \\ &= 3x^2 \end{aligned}$$

$$\underline{f_2'(x) = 3x^2_{\#}}$$

$$f_3(x) = x \cdot x^3$$

$$\begin{aligned} f_3'(x) &= (x)' \cdot x^3 + x \cdot (x^3)' \\ &= x^3 + x \cdot 3x^2 \\ &= 4x^3 \end{aligned}$$

$$\underline{f_3'(x) = 4x^3_{\#}}$$

(PA 4.4)

$$\frac{d}{dx} f(g(x)) = \frac{df(u)}{du} \cdot \frac{du}{dx} \quad (u = g(x))$$

$$f_1(x) = (x+1)^3$$

$$\begin{aligned} f_1'(x) &= \frac{d(x+1)^3}{(x+1)} \cdot \frac{d(x+1)}{dx} \\ &= 3(x+1)^2 \cdot 1 \end{aligned}$$

$$= 3(x+1)^2$$

$$\underline{f_1'(x) = 3(x+1)^2_{\#}}$$

$$f_2(x) = (4x)^3$$

$$f_2'(x) = \frac{d(4x)^3}{d(4x)} \cdot \frac{d(4x)}{dx}$$

$$= 3(4x)^2 \cdot 4$$

$$= 12 \cdot (4x)^2$$

$$\underline{f_2'(x) = 12(4x)^2}_{//}$$

$$f_3(x) = \exp(2x-1)$$

$$f_3'(x) = \frac{d \exp(2x-1)}{d(2x-1)} \cdot \frac{d(2x-1)}{dx}$$

$$= \exp(2x-1) \cdot 2$$

$$\underline{f_3'(x) = 2 \exp(2x-1)}_{//}$$

$$f_4(x) = \exp(-x^2)$$

$$f_4'(x) = \frac{d \exp(-x^2)}{d(-x^2)} \cdot \frac{d(-x^2)}{dx}$$

$$= \exp(-x^2) \cdot (-2x)$$

$$\underline{f_4'(x) = -2x \exp(-x^2)}_{//}$$

$$f_5(x) = 2^x \quad [= \exp(\log(2)x)]$$

$$f_5'(x) = \frac{d \exp(\log(2)x)}{d(\log(2)x)} \cdot \frac{d(\log(2)x)}{dx}$$

$$= \exp(\log(2)x) \cdot \log(2)$$

$$\underline{f_5'(x) = \log(2) 2^x}_{//}$$

(問 4.5) $f(x) = x^2 + 1$, $g(x) = f(f(x))$

1. (証明)

$$\begin{aligned} g(x) &= f(f(x)) \\ &= f(x^2 + 1) \\ &= (x^2 + 1)^2 + 1 \\ &= x^4 + 2x^2 + 1 + 1 \\ &= x^4 + 2x^2 + 2 \quad (\text{終}) \end{aligned}$$

(証明)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4x^3h + 4x^2h^2 + 4xh^3 + h^4 + 4xh + 2h^2}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0} (4x^3 + 4x^2h + 4xh^2 + h^3 + 4x + 2h)$$

$$= 4x^3 + 4x$$

$$g'(x) = 4x^3 + 4x \quad (\text{終})$$

$$\begin{aligned} g(x+h) &= (x+h)^4 + 2(x+h)^2 + 2 \\ &= (x+h)^2 \{(x+h)^2 + 2\} + 2 \\ &= (x^2 + 2xh + h^2) \\ &\quad (x^2 + 2xh + h^2 + 2) + 2 \\ &= x^4 + 2x^3h + x^2h^2 + 2x^2 \\ &\quad + 2x^3h + 2x^2h^2 + 2xh^3 \\ &\quad + 4xh + 2x^2h^2 + 2xh^3 + h^4 \\ &\quad + 2h^2 + 2 \\ &= x^4 + 4x^3h + 4x^2h^2 \\ &\quad + 4xh^3 + h^4 \\ &\quad + 2x^2 + 4xh + 2h^2 + 2 \end{aligned}$$

2. (証明)

$$g'(x) = \frac{d}{dx} f(f(x))$$

$$= \frac{df(f(x))}{df(x)} \cdot \frac{df(x)}{dx}$$

$$= \frac{d(f(x)^2 + 1)}{df(x)} \cdot \frac{d(x^2 + 1)}{dx}$$

$$= 2f(x) \cdot 2x$$

$$= 4x(x^2 + 1)$$

$$= 4x^3 + 4x \quad (\text{終})$$

$$3. \quad h(x) = f(f(f(x)))$$

$$g(x) = f(f(x)) \quad f'$$

$$h(x) = f(g(x))$$

$$\frac{dh(x)}{dx} = \frac{df(g(x))}{dx}$$

$$= \frac{df(g(x))}{dg(x)} \cdot \frac{dg(x)}{dx}$$

$$= \frac{d(g(x)^2 + 1)}{dg(x)} \cdot \frac{d(x^4 + 2x^2 + 2)}{dx}$$

$$= 2g(x) \cdot (4x^3 + 4x)$$

$$= 2(x^4 + 2x^2 + 2)(4x^3 + 4x)$$

$$= 2(4x^7 + 4x^5 + 8x^5 + 8x^3 + 8x^3 + 8x)$$

$$h'(x) = 8x^7 + 8x^5 + 16x^5 + 16x^3 + 16x^3 + 16x //$$

(Pg 4.7)

$$\cdot \text{ReLU}(x) = \max(x, 0)$$

$$\frac{d\text{ReLU}(x)}{dx} = \lim_{h \rightarrow 0} \frac{\text{ReLU}(x+h) - \text{ReLU}(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\max(x+h, 0) - \max(x, 0)}{h} \quad \text{--- (1)}$$

$$x+h > 0$$

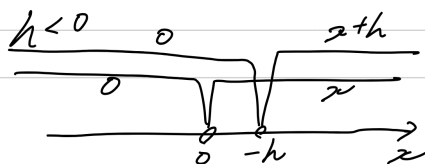
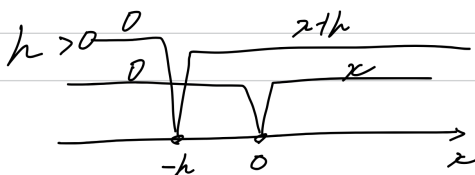
$$x > 0$$

$$x > -h$$

$$x < 0$$

$$x+h < 0$$

$$x < -h$$



[1] $h > 0$ の $x \pm$,

(i) $x < -h$ の $x \pm$,

$$\lim_{h \rightarrow 0} \frac{0 - 0}{h} = \underline{0}$$

(ii) $-h < x < 0$ の $x \pm$,

$$\lim_{h \rightarrow 0} \frac{x+h+0}{h} = \lim_{h \rightarrow 0} \left(\frac{x}{h} + 1 \right) = \underline{\infty}$$

(iii) $0 < x$ の $x \pm$,

$$\lim_{h \rightarrow 0} \frac{x+h-x}{h} = \lim_{h \rightarrow 0} 1 = \underline{1}$$

[2] $h < 0$ の $x \pm$,

(i) $x < 0$ の $x \pm$,

$$\lim_{h \rightarrow 0} \frac{0 - 0}{h} = \underline{0}$$

(ii) $0 < x < -h$ の $x \pm$ ~~∞~~

(iii) $x > -h$ の $x \pm$, 1

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\begin{aligned} \tanh'(x) &= \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} \\ &= \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} - 2 + e^{-2x})}{e^{2x} + 2 + e^{-2x}} \\ &= \frac{4}{e^{2x} + e^{-2x} + 2} \end{aligned}$$

$$\text{Sigmoid}(x) = \frac{1}{1 + e^x}$$

$$\text{Sigmoid}'(x) = \frac{-(1 - e^{-x})}{(1 + e^{-x})^2}$$

$$\text{Softplus}(x) = \log(1 + e^x)$$

$$\begin{aligned} \text{Softplus}'(x) &= \frac{1}{1 + e^x} (e^x) \\ &= \frac{e^x}{1 + e^x} \end{aligned}$$

(PA 4.8)

$$1. f(x, y) = |x| + |y|$$

$$(i) x > 0 \text{ のとき, } f(x, y) = x + |y|$$

$$\frac{\partial}{\partial x} f(x, y) = 1$$

$$(ii) x < 0 \text{ のとき, } f(x, y) = -x + |y|$$

$$\frac{\partial}{\partial x} f(x, y) = -1$$

$$\frac{\partial}{\partial x} f(x, y) = \begin{cases} 1 & (x > 0) \\ -1 & (x < 0) \end{cases}$$

$$(iii) y > 0 \text{ のとき, } f(x, y) = |x| + y$$

$$\frac{\partial}{\partial y} f(x, y) = 1$$

$$(iv) y < 0 \text{ のとき, } f(x, y) = |x| - y$$

$$\frac{\partial}{\partial y} f(x, y) = -1$$

$$\frac{\partial}{\partial y} f(x, y) = \begin{cases} 1 & (y > 0) \\ -1 & (y < 0) \end{cases}$$

$$2. f(x, y) = |x + y|$$

$$(i) x + y > 0 \text{ の } \forall x, y, f(x, y) = x + y$$

$$\frac{\partial}{\partial x} f(x, y) = 1, \quad \frac{\partial}{\partial y} f(x, y) = 1$$

$$(ii) x + y < 0 \text{ の } \forall x, y, f(x, y) = -(x + y) = -x - y$$

$$\frac{\partial}{\partial x} f(x, y) = -1, \quad \frac{\partial}{\partial y} f(x, y) = -1$$

$$\frac{\partial}{\partial x} f(x, y) = \begin{cases} 1 & (x > -y) \\ -1 & (x < -y) \end{cases}$$

$$\frac{\partial}{\partial y} f(x, y) = \begin{cases} 1 & (x > -y) \\ -1 & (x < -y) \end{cases}$$

//

(問5.2)

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, \frac{\partial}{\partial x} (x^T A x) = x^T (A + A^T)$$

$$1. x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \text{ であることを示す。}$$

$$\pm \text{ 5.1-1, } x^T A x = a_{11} x_1^2 + \left(\sum_{i=2}^n (a_{i1} + a_{1i}) x_i \right) + \sum_{i=2}^n \sum_{j=2}^n x_i x_j a_{ij} \text{ を示す。}$$

(証明)

$$\text{すなわち, } x^T = [x_1 \cdots x_n]$$

$$x^T A x = [x_1 \cdots x_n] \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$= [x_1 \cdots x_n] \begin{bmatrix} a_{11} x_1 + \cdots + a_{1n} x_n \\ \vdots \\ a_{n1} x_1 + \cdots + a_{nn} x_n \end{bmatrix}$$

$$\begin{aligned}
&= x_1(a_{11}x_1 + \dots + a_{1n}x_n) \\
&\quad + \dots + x_n(a_{n1}x_1 + \dots + a_{nn}x_n) \\
&= x_1 \left(\sum_{j=1}^n a_{1j}x_j \right) + \dots + x_n \left(\sum_{j=1}^n a_{nj}x_j \right) \\
&= \sum_{i=1}^n \left(x_i \sum_{j=1}^n a_{ij}x_j \right) \\
&= \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_ix_j \quad (\text{終})
\end{aligned}$$

(証明)

$$x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

$$\begin{aligned}
&= \sum_{j=1}^n a_{1j} x_1 x_j + \sum_{i=2}^n \sum_{j=1}^n a_{ij} x_i x_j \\
&= a_{11}x_1x_1 + \sum_{j=2}^n a_{1j}x_1x_j + \sum_{i=2}^n \sum_{j=1}^n a_{ij}x_ix_j \\
&= a_{11}x_1^2 + \sum_{j=2}^n a_{1j}x_1x_j + \sum_{i=2}^n a_{i1}x_ix_1 \\
&\quad + \sum_{i=2}^n \sum_{j=2}^n a_{ij}x_ix_j \\
&= a_{11}x_1^2 + \left(\sum_{i=2}^n (a_{1i} \underline{x_1 x_i} + a_{i1} \underline{x_i x_1}) \right) \\
&\quad + \sum_{i=2}^n \sum_{j=2}^n a_{ij}x_ix_j \\
&= a_{11}x_1^2 + \left(\sum_{i=2}^n (a_{1i} + a_{i1})x_i \right) x_1 + \sum_{i=2}^n \sum_{j=2}^n x_ix_j a_{ij} \quad (\text{終})
\end{aligned}$$

$$2. \frac{\partial}{\partial x_1} (x^T A x) \text{ を計算し, } \sum_{i=1}^n (a_{1i} x_i + a_{i1} x_i) \text{ とおす} \\ = 2 \text{ と示す。}$$

(証明)

$$x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \\ = a_{11} x_1^2 + \left(\sum_{i=2}^n (a_{1i} + a_{i1}) x_i \right) x_1 + \sum_{i=2}^n \sum_{j=2}^n x_i x_j a_{ij}$$

$$\frac{\partial}{\partial x_1} (x^T A x) = 2a_{11} x_1 + \sum_{i=2}^n (a_{1i} + a_{i1}) x_i \\ = (a_{11} + a_{11}) x_1 + \sum_{i=2}^n (a_{1i} + a_{i1}) x_i \\ = \sum_{i=1}^n (a_{1i} + a_{i1}) x_i \quad (\text{終})$$

$$3. \frac{\partial}{\partial x_k} (x^T A x) = \sum_{i=1}^n (a_{ki} x_i + a_{ik} x_i) \text{ とおす} = 2 \text{ と示す。}$$

$$\text{また, } \frac{\partial}{\partial x} (x^T A x) = x^T (A + A^T) \text{ と確かめる。}$$

(証明)

$$\frac{\partial}{\partial x_k} (x^T A x) = \sum_{i=1}^n (a_{ki} x_i + a_{ik} x_i) \text{ とおす。}$$

$= 2x_i, k=1 \text{ のとき}$

$$\frac{\partial}{\partial x_1} (x^T A x) = \sum_{i=1}^n (a_{1i} + a_{i1}) x_i \text{ となる。} \quad \textcircled{1}$$





次に、 $k = k+1$ を考える。

$$\begin{aligned} x^T A x &= \sum_{\bar{i}=1}^n \sum_{j=1}^n a_{\bar{i}j} x_{\bar{i}} x_j \\ &= \sum_{\bar{i}=1}^n \sum_{j=1}^k a_{\bar{i}j} x_{\bar{i}} x_j + \sum_{\bar{i}=1}^n a_{\bar{i}(k+1)} x_{\bar{i}} x_{(k+1)} \\ &\quad + \sum_{\bar{i}=1}^n \sum_{j=k+2}^n a_{\bar{i}j} x_{\bar{i}} x_j \end{aligned}$$

$$\begin{aligned} &= \sum_{\bar{i}=1}^k \sum_{j=1}^k a_{\bar{i}j} x_{\bar{i}} x_j + \sum_{j=1}^k a_{(k+1)j} x_{(k+1)} x_j + \sum_{\bar{i}=k+2}^n a_{\bar{i}j} x_{\bar{i}} x_j \\ &\quad + \sum_{\bar{i}=1}^k a_{\bar{i}(k+1)} x_{\bar{i}} x_{(k+1)} + a_{(k+1)(k+1)} x_{(k+1)} x_{(k+1)} + \sum_{\bar{i}=k+2}^n a_{\bar{i}(k+1)} x_{\bar{i}} x_{(k+1)} \\ &\quad + \sum_{\bar{i}=1}^k \sum_{j=k+2}^n a_{\bar{i}j} x_{\bar{i}} x_j + \sum_{j=k+2}^n a_{(k+1)j} x_{(k+1)} x_j + \sum_{\bar{i}=k+2}^n \sum_{j=k+2}^n a_{\bar{i}j} x_{\bar{i}} x_j \\ \frac{\partial}{\partial x_{(k+1)}} (x^T A x) &= \sum_{j=1}^k a_{(k+1)j} x_j + \sum_{\bar{i}=1}^k a_{\bar{i}(k+1)} x_{\bar{i}} + 2 a_{(k+1)(k+1)} x_{(k+1)} \\ &\quad + \sum_{\bar{i}=k+2}^n a_{\bar{i}(k+1)} x_{\bar{i}} + \sum_{j=k+2}^n a_{(k+1)j} x_j \\ &= \sum_{j=1}^k a_{(k+1)j} x_j + a_{(k+1)(k+1)} x_{(k+1)} + \sum_{j=k+2}^n a_{(k+1)j} x_j \\ &\quad + \sum_{\bar{i}=1}^k a_{\bar{i}(k+1)} x_{\bar{i}} + a_{(k+1)(k+1)} x_{(k+1)} + \sum_{\bar{i}=k+2}^n a_{\bar{i}(k+1)} x_{\bar{i}} \\ &= \sum_{j=1}^n a_{(k+1)j} x_j + \sum_{\bar{i}=1}^n a_{\bar{i}(k+1)} x_{\bar{i}} \\ &= \sum_{\bar{i}=1}^n (a_{(k+1)\bar{i}} x_{\bar{i}} + a_{\bar{i}(k+1)} x_{\bar{i}}) \end{aligned}$$

以上のことから、 $\frac{\partial}{\partial x_k} (x^T A x) = \sum_{\bar{i}=1}^n (a_{k\bar{i}} x_{\bar{i}} + a_{\bar{i}k} x_{\bar{i}})$

が成り立つ。

(証明)

$$\frac{\partial}{\partial x_k} (x^T A x) = \sum_{i=1}^n (a_{ki} x_i + a_{ik} x_i)$$

$$\begin{aligned} &= a_{k1} x_1 + a_{1k} x_1 + a_{k2} x_2 + a_{2k} x_2 \\ &+ a_{k3} x_3 + a_{3k} x_3 + a_{k4} x_4 + a_{4k} x_4 \\ &\quad \vdots \\ &+ a_{kn} x_n + a_{nk} x_n \end{aligned}$$

$$= [x_1 \cdots x_n] \begin{bmatrix} a_{k1} \\ \vdots \\ a_{kn} \end{bmatrix} + [x_1 \cdots x_n] \begin{bmatrix} a_{1k} \\ \vdots \\ a_{nk} \end{bmatrix}$$

つまり、

$$\frac{\partial}{\partial x} (x^T A x) = [x_1 \cdots x_n] \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

$$+ [x_1 \cdots x_n] \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

$$\therefore \text{つまり、} A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, A^T = \begin{bmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & & \vdots \\ a_{1n} & \cdots & a_{nn} \end{bmatrix} \text{ である。}$$

$$\frac{\partial}{\partial x} (x^T A x) = x^T A^T + x^T A$$

$$= x^T (A + A^T) \quad (\text{終})$$

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